

Weather Impact Analysis On Crops

Dashboard



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**A TECHNICAL REPORT SUBMITTED IN PARTIAL FULFILLMENT OF
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BACHERLORS OF SCIENCE

IN

SOFTWARE ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF SCIENCES

UNIVERSITY OF AGRICULTURE FAISALABAD

DECLARATION

I hereby declare that the contents of the report **Weather impact analysis on crops dashboard** represent my own research and work, and no part has been copied from any published source except for the references duly cited. I further declare that this work has not been submitted for the award of any other diploma or degree. The University of Agriculture, Faisalabad, may take action if the information provided is found false at any stage. In case of any default, I shall be proceeded against as per UAF policy

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Project Evaluation Matrix (Total: 80 Marks)

Category	Obtained Marks	Max Marks
Project Documentation		15
Originality & Code Quality		10
Features Developed		20
User Interface (UI/UX)		10
Portability & Deployment		10
Technical Knowledge (Viva)		15

CERTIFICATE

To,
The Controller of Examinations,
University of Agriculture,
Faisalabad.

The supervisory committee certify that **Masooma Khatoon 2022-ag-8011 & Abiha Khalil 2022-ag-8006** has successfully completed his project in partial fulfillment of requirement for the degree of BS Software Engineering under our guidance and supervision.

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ABSTRACT

This project, titled **Weather Impact Analysis on Crops Dashboard**, aims to analyze the influence of weather conditions on crop growth and agricultural productivity. A web-based dashboard was developed using modern technologies to collect, process, and visualize weather and crop data in an interactive and user-friendly manner. The system integrates historical and real-time weather parameters such as temperature, rainfall, humidity, and wind speed to identify patterns and correlations with crop yield and health.

The dashboard provides features for farmers, researchers, and agricultural experts to monitor environmental conditions, compare crop performance, and generate insights for better decision-making. Data visualization techniques, including charts and graphs, are used to present trends and predictive analysis clearly. The system was evaluated using agricultural and meteorological datasets, demonstrating accurate analysis and meaningful insights into the effects of weather on crop production.

The proposed solution improves agricultural planning by helping users anticipate risks, optimize irrigation and fertilization schedules, and select suitable crops based on climatic conditions. Overall, this project highlights how data analytics and dashboard technologies can be used to support smart farming and enhance agricultural productivity efficiently.

GitHub Link :

<https://github.com/Masooma22/Weather-Impact-Analysis-on-Crops-Dashboard>

<https://github.com/biyaa-8006/Agroinsight.git>

Table of Contents

I hereby declare that the contents of the report Weather impact analysis on crops dashboard

Chapter 1 - INTRODUCTION	1
1.1 Background:	1
1.2 Description:	1
1.3 Problem Statement:	1
1.4 Scope:	2
1.5 Objectives:	2
1.6 Feasibility:	2
1.7.1 Functional Requirements	4
FR01: Provide user name and password to log in	4
1.7.2 Non- Functional Requirements	6
1.7.3 Hardware Requirements	6
End-User Devices (Web Dashboard Access via Browser)	6
● Development and Testing Environment	7
1.7.4 Software Requirements	8
1.7 Stakeholders:	8
• Farmers	9
• Researchers and Agricultural Experts	9
• Administrators	9
Secondary Stakeholders	9
• Project Sponsor / Management Team	9
• Technical Development Team	9
Tertiary Stakeholders	10
• Hosting Providers / Third-Party Service Providers	10
Chapter 2 – MATERIALS & METHODS	11
2.1 Process Model:	11
Requirement Analysis Phase	11
Design Phase	11
Development Phase	12
Testing Phase	12
Deployment Phase	12
Advantages of Agile Methodology for This Project	13

2.2 Tools & Technologies	14
Frontend Development	14
1. HTML5	14
2. CSS3	14
3. JavaScript (ES6+)	14
4. Chart.js / Plotly	14
Backend Development	14
1. Python	14
3. MySQL	15
4. Pandas	15
5. NumPy	15
6. Scikit-learn	15
2.3 Design	15
2.3.1 Use Case Diagrams	16
Actors:	16
Main Use Cases (Based on Weather Impact Analysis on Crops Dashboard Features):	16
Table 2. 1: Register User	18
Table 2. 2: Login	18
Table 2. 3: Upload Dataset	18
Table 2. 4: Generate Crop Recommendations	19
2.3.3 Sequence Diagram:	19
2.3.4 Class Diagram:	20
2.3.4 Data Flow Diagram :	20
.....	21
Figure 2.3 : DFD	21
2.3.6 ER Diagram:	22
2.3.7 Database Model:	22
Figure 2.8 Database Model 1	23
2.3.7 Architecture:	23
Presentation Tier	23
Application Tier	24
Data Tier	24
Chapter 3 - RESULTS & DISCUSSION	26
3.1 Testing:	26

1. Unit Testing	26
2. Integration Testing	26
3. System Testing	27
4. Performance Testing	27
5. Usability Testing	28
Testing Coverage	28
Best Practices for Test Cases	28
3.2 Test Cases:	29
Test Case 1 – User Registration	29
Test Case 2 – User Login	29
Test Case 3 – Dataset Upload	29
Test Case 4 – Weather Trend Analysis	30
Test Case 5 – Crop Yield Comparison	30
Test Case 6 – Recommendation Generation	30
Test Case 7 – Report Generation	31
Test Case 8 – Notification System	31
Test Case 9 – User Profile Management	31
Test Case 10 – Historical Analysis Access	32
Test Case 11 – Admin Dashboard Management	32
3.3 Conclusion:	32
Chapter 4 - USER MANUAL	34

List of Figures

<i>Figure 2.1 Agile Activities</i>	13
<i>Figure 2.2 Use Case Diagram</i>	17
<i>Figure 2.1 Sequence Diagram</i>	19
<i>Figure 2.2 Class Diagram</i>	20

List of Tables

Table 2.1 User Registration.....	20
Table 2.2: User Login.....	20
Table 2.3 Dataset Upload.....	20
Table 2.4: Weather Trend Analysis.....	20
Table 3.1: Crop Yield Comparison.....	23
Table 3.2: Recommendation Generation.....	24
Table 3.3: Report Generation.....	25
Table 3.4 Appointment Booking Test Case.....	27
Table 3.5 User Dashboard Test Case.....	28
Table 3.6 Alert Update Test Case	28
Table 3.7 Economic impact Test Case.....	28
Table 3.8 Prediction Test Case.....	29
Table 3.9: Logout Functionality Test Case.....	29
Table 3.10: Historical Analysis Access.....	29
Table 3.11 User Profile Management Test.....	29
Table 3.12 Notification System Test Case.....	30
Table 3.13: Admin Dashboard Management Test.....	30

Chapter 1 - INTRODUCTION

1.1 Background:

The Weather Impact Analysis on Crops Dashboard is a web-based agricultural platform designed to help farmers, researchers, and agricultural experts analyze the effects of weather conditions on crop production. The system digitalizes agricultural data analysis by providing interactive dashboards for monitoring temperature, rainfall, humidity, wind speed, and crop performance. Users can upload datasets, visualize trends, compare crop yields, and access predictive insights for better decision-making. Developed using Php, the dashboard reduces manual effort, improves data management, and provides secure access to weather and crop information. By simplifying complex agricultural data analysis, the system supports more efficient, reliable, and informed agricultural planning.

1.2 Description:

The Weather Impact Analysis on Crops Dashboard is a web-based system developed to help farmers, researchers, and agricultural experts analyze the relationship between weather conditions and crop productivity. The platform enables users to upload datasets, monitor weather parameters, visualize crop performance, and generate predictive insights through interactive dashboards. It reduces the challenges of manual data analysis, improves data organization, and supports informed agricultural decision-making. Built using Python with MySQL for secure data storage, the system provides features such as weather trend analysis, crop comparison, report generation, and personalized dashboards. By offering accurate and accessible agricultural insights, the application enhances crop management, planning efficiency, and overall productivity.

1.3 Problem Statement:

The Weather Impact Analysis on Crops Dashboard is a secure web-based platform designed to help farmers, researchers, and agricultural experts analyze the effects of weather conditions on crop productivity. The system provides interactive dashboards for weather trend visualization, crop performance monitoring, predictive insights, and report generation. It reduces the

limitations of traditional manual methods by improving data organization, eliminating repetitive calculations, and simplifying agricultural analysis. With secure data storage and easy access to weather and crop information, the platform supports informed decision-making, efficient agricultural planning, and better crop management, making farming more productive and data-driven.

1.4 Scope:

The Weather Impact Analysis on Crops Dashboard is a web-based platform that helps farmers, researchers, and agricultural organizations analyze the effects of weather on crop productivity. The system enables users to upload datasets, monitor weather trends, compare crop yields, access predictive insights, and generate reports through personalized dashboards. It provides analytical tools for evaluating factors such as temperature, rainfall, humidity, wind speed, and soil moisture. Using MySQL for secure data storage, the platform ensures accurate, accessible, and reliable management of weather and crop information. The system improves agricultural planning, reduces manual analysis, and supports better decision-making through a secure and user-friendly environment.

1.5 Objectives:

The main objective of the Weather Impact Analysis on Crops Dashboard is to provide a secure and user-friendly web platform that helps farmers, researchers, and agricultural organizations analyze weather and crop data efficiently. The system enables users to visualize weather trends, compare crop performance, generate reports, and access predictive insights. By monitoring factors such as temperature, rainfall, humidity, and wind speed, the platform supports better agricultural planning and informed decision-making. Developed using Php, the dashboard reduces manual analysis, improves data accessibility, and makes agricultural insights more organized, accurate, and efficient.

1.6 Feasibility:

The feasibility study for the Weather Impact Analysis on Crops Dashboard evaluates its viability across technical, schedule, economic, cultural, legal/ethical, resource, and operational

dimensions to ensure a successful and sustainable digital agricultural analytics solution. This comprehensive analysis validates the system's practicality, user-centered design, scalability, and compliance with data management standards, ensuring that the solution aligns with agricultural requirements and project constraints.

Technical Feasibility: The project is technically sound and achievable, leveraging the development team's proficiency in web technologies such as Php (frontend), MySQL (database), and JavaScript (programming language), with integration support for data upload, weather analysis, chart generation, and real-time dashboard updates. The application is built with a modular architecture, allowing secure user dashboards, weather trend visualization, crop yield comparison, predictive analysis, and report generation. Development using Visual Studio Code ensures efficient implementation and cross-platform accessibility through modern web browsers. The selected technology stack is modern, maintainable, and aligned with current web application development standards.

Schedule Feasibility: The project is structured for completion within a 16–18 week timeline, utilizing Agile methodology with defined sprints for requirement gathering, frontend and backend development, feature integration, testing, and deployment. Milestones such as dataset upload and preprocessing implementation (by week 5), dashboard visualization and analytical features (by week 10), and complete system testing with optimization (by week 14) ensure that major functionalities are developed incrementally. Regular feedback from supervisors, testers, and users ensures timely improvements and alignment with agricultural analytics requirements.

Economic Feasibility: The system is economically viable, relying on cost-effective and open-source technologies such as Php, JavaScript, MySQL, and Visual Studio Code. The use of open-source tools reduces development and deployment costs, making the solution affordable for agricultural institutions and research organizations. Development costs are minimized because of freely available frameworks and tools, making the project budget-friendly while still delivering efficient agricultural analytics services.

Cultural Feasibility: The Weather Impact Analysis on Crops Dashboard is designed to be user-friendly and accessible for farmers, researchers, and agricultural experts from different

educational and cultural backgrounds. The application reduces information barriers by providing clear visualizations of weather trends, crop performance, and predictive insights in a simple and organized manner. By replacing manual analysis with a secure digital system, the application improves trust, convenience, and accessibility for users in both academic and practical farming environments.

Legal/Ethical Feasibility: The project follows legal and ethical standards related to secure data management and responsible use of agricultural information. Weather and crop datasets are stored securely in MySQL databases. User access is protected through secure login systems, ensuring that sensitive data and reports are accessible only to authorized users. Ethical considerations include accurate data representation, transparency in analytical results, and responsible handling of agricultural recommendations.

Operational Feasibility: After deployment, the application is designed to operate efficiently with minimal manual intervention. Farmers, researchers, and agricultural experts are provided with intuitive dashboards that require little technical knowledge to use effectively. Features such as dataset upload, weather trend analysis, crop comparison, predictive insights, and report generation are streamlined through responsive web interfaces. Administrative tasks related to data management and reporting are simplified, reducing workload and improving decision-making efficiency. System maintainability is ensured through modular development practices, proper documentation, and scalable database services, allowing future expansion and feature enhancements over time.

1.7.1 Functional Requirements

Following are some all functional requirements of my system

FR01: Provide user name and password to log in

FR01-01	System shall get Username and Password from user
FR01-02	System should authenticate user name and password
FR01-03	System shall let the user to log in if information is valid
FR01-04	If information is not valid then system will display message to get the account by admin

FR02: User Roles

FR02-01	System shall provide Admin role
FR02-02	System shall provide Farmer/Client role
FR02-03	System shall restrict access according to role privileges

FR03: Weather Data Collection

FR03-01	System shall collect real-time weather data (temperature, humidity, rainfall, wind)
FR03-02	System shall retrieve historical weather data
FR03-03	System shall store weather data in database

FR04: Crop Selection

FR04-01	System shall allow users to select crop type (e.g., Wheat, Rice, Cotton, etc.)
FR04-02	System shall allow users to select geographic region/location

FR05: Weather Impact Analysis

FR05-01	System shall analyze collected weather data
FR05-02	System shall evaluate impact of weather on selected crop
FR05-03	System shall calculate risk level (Low/Medium/High)

FR06: Recommendation System

FR06-01	System shall provide suggestions to minimize risk
FR06-02	System shall provide irrigation and fertilizer recommendations
FR06-03	System shall suggest best crop growing period

FR07: Alerts & Notification

FR07-01	System shall notify users about extreme weather condition
FR07-02	System shall send alerts via dashboard notifications
FR07-03	System shall send reminders for important crop activities

FR08: Reports & Visualization

FR08-01	System shall generate weather impact reports
FR08-02	System shall display graphical charts and dashboards
FR08-03	System shall allow users to download reports

FR09: Data Management

FR09-01	System shall store user data securely
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FR09-02	System shall allow admin to update/delete records
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FR10: System Administration

FR10-01	Admin shall manage users
FR10-02	Admin shall manage crops database
FR10-03	Admin shall manage system configuration

1.7.2 Non- Functional Requirements

Following are some all non-functional requirements of my system

NFR01: System shall remain available **24/7**.

NFR02: System shall display dashboard results within **3 seconds**.

NFR03: System shall provide accurate data with **95% reliability**.

NFR04: System shall support an increasing number of users and locations.

NFR05: System shall encrypt user credentials. System shall restrict access based on roles.

NFR06: System shall provide a user-friendly and simple interface.

NFR07: System shall allow easy updates and modifications.

NFR08: System shall ensure no data loss occurs during processing.

NFR09: System shall integrate with weather APIs and external services.

NFR10: Weather and analysis results should be **highly accurate and validated**..

1.7.3 Hardware Requirements

The Weather Impact Analysis on Crops Dashboard, developed using Php in Visual Studio Code, requires specific hardware capabilities to ensure smooth performance for end users (farmers, researchers, and agricultural experts) as well as the development and testing environment. The following outlines the minimum and recommended hardware specifications.

End-User Devices (Web Dashboard Access via Browser)

Minimum Requirements:

- **Processor:** Dual-core processor (e.g., Intel Core i3, AMD Ryzen 3, or equivalent).
- **Memory:** 2 GB RAM for basic dashboard usage and smooth navigation.
- **Storage:** 200 MB free space for browser cache and temporary files.
- **Network:** Stable internet connection (Wi-Fi or 4G) for uploading datasets and viewing interactive charts.
- **Display:** 720p resolution (1280×720) for clear visualization of graphs and dashboard components.
- **Browser:** Latest version of Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari.

Recommended Requirements:

- **Processor:** Quad-core processor or higher (e.g., Intel Core i5, AMD Ryzen 5, or equivalent).
- **Memory:** 4 GB RAM or more for smooth multitasking and handling larger datasets.
- **Storage:** 1 GB free space for cached datasets, exported reports, and browser data.
- **Network:** High-speed internet connection (Wi-Fi or 5G) for faster data upload and real-time dashboard updates.
- **Display:** Full HD resolution (1920×1080) or higher for better chart and dashboard visibility.
- **Browser:** Latest version of Google Chrome or Microsoft Edge for optimal performance.
- **Development and Testing Environment**

Minimum Requirements:

- **Processor:** Intel Core i3 or equivalent.
- **Memory:** 4 GB RAM.
- **Storage:** 10 GB free disk space.
- **Operating System:** Windows 10, Linux, or macOS.
- **Software:** Visual Studio Code, Node.js, MySQL, and modern web browsers

Recommended Requirements:

- **Processor:** Intel Core i5 or higher.
- **Memory:** 8 GB RAM or more.
- **Storage:** 20 GB free disk space.
- **Operating System:** Windows 11, Ubuntu 22.04, or macOS.
- **Software:** Visual Studio Code, Node.js, MySQL, Git, and browser developer tools.

1.7.4 Software Requirements

End-User Environment (Mobile Application – Expo Go / React Native App):

- **Supported Platforms:**
 - Android devices running Android 9.0 (Pie) or later
 - iOS devices running iOS 13 or later
 - Expo Go application (latest version) for testing and runtime
- **Development Environment:**
 - Visual Studio / Visual Studio Code
 - Node.js (LTS version)
 - Expo CLI (latest version)
 - Firebase (for backend services and database)
- **Operating Systems (Development):**
 - Windows 10 or later
 - macOS 10.14 (Mojave) or later
 - Linux (Ubuntu 18.04 or later)

1.7 Stakeholders:

Stakeholders are individuals or entities with a vested interest in the success, usability, and performance of the Weather Impact Analysis on Crops Dashboard. Understanding and fulfilling their requirements is essential for effective system design, smooth functionality, and long-term sustainability.

Primary Stakeholders

- **Farmers**

End-users who access the dashboard to upload agricultural and meteorological datasets, monitor weather trends, analyze crop performance, view predictive insights, and generate reports. Their usage and feedback directly determine the system's success and practical value in improving farming decisions.

- **Researchers and Agricultural Experts**

Agricultural scientists and domain experts who analyze the relationship between weather conditions and crop productivity. They use the dashboard to study historical and real-time data, compare crop yields, generate recommendations, and validate analytical models. Their expertise ensures accurate interpretation of results and meaningful agricultural insights.

- **Administrators**

System managers responsible for user account management, dataset maintenance, report oversight, system monitoring, and handling technical issues. They ensure smooth operation, data accuracy, security, and overall reliability of the dashboard.

Secondary Stakeholders

- **Project Sponsor / Management Team**

Responsible for overseeing the project's objectives, timeline, and budget. They ensure that the Weather Impact Analysis on Crops Dashboard aligns with academic, research, or organizational goals and provide the necessary guidance and resources.

- **Technical Development Team**

Includes developers, UI/UX designers, database administrators, and QA testers who design, develop, and test the application using Php, MySQL, and Visual Studio Code. Their work ensures the system is functional, reliable, scalable, and user-friendly.

Tertiary Stakeholders

• Hosting Providers / Third-Party Service Providers

Provide essential infrastructure and services such as web hosting, cloud storage, and data APIs for weather and agricultural datasets. Their reliability and performance directly support the scalability, availability, and smooth functioning of the Weather Impact Analysis on Crops Dashboard.

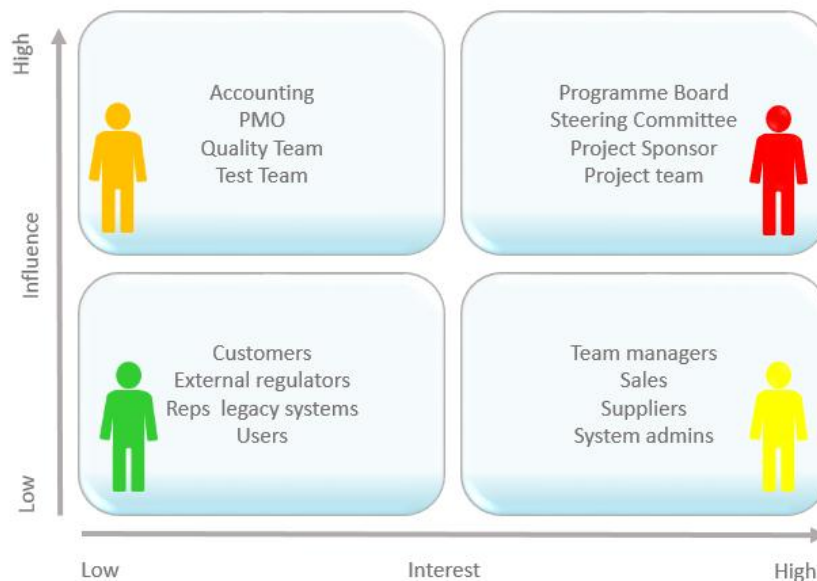


Figure 1.1 Stakeholders

Chapter 2 – MATERIALS & METHODS

2.1 Process Model:

The Weather Impact Analysis on Crops Dashboard follows the Agile development methodology, which supports flexible and iterative development through short sprints. Features such as dataset upload, weather analysis, crop comparison, predictive insights, and report generation are developed and tested incrementally. Regular reviews and stakeholder feedback ensure continuous improvement, enhanced functionality, and a user-friendly system.

Planning Phase

The project originated from the need for an efficient and digital agricultural analytics system that helps users understand how weather conditions affect crop growth and productivity. The idea was developed by analyzing common challenges such as scattered weather records, manual data analysis, inaccurate predictions, and limited access to timely agricultural insights. Consultations with farmers, researchers, and agricultural experts helped define the system vision: a web-based dashboard that integrates weather and crop data, visualizes trends, and provides predictive insights and recommendations. Key goals included improving agricultural decision-making, reducing weather-related risks, and ensuring smooth management of analytical data and reports.

Requirement Analysis Phase

Detailed discussions with stakeholders—including farmers, researchers, and administrators—guided the identification of both functional requirements (e.g., dataset upload, weather visualization, crop yield comparison, predictive analysis, and report generation) and non-functional requirements (e.g., performance, security, usability, and scalability). The analysis also focused on role-based access control, secure handling of agricultural data, and efficient processing of large datasets. All requirements were documented to ensure alignment with system objectives and to support future enhancements.

Design Phase

System architecture and UI prototypes were developed during this phase. Wireframes and mockups were created for key modules such as the user dashboard, data upload interface, weather analysis charts, crop comparison section, and reporting module. The interface was designed to be simple, intuitive, and responsive for smooth usage on desktops, laptops, and mobile browsers. Feedback from early users and agricultural experts was used to improve UI/UX, ensuring easy navigation, clarity, and accessibility for users with different technical backgrounds.

Development Phase

The development was carried out in iterative sprint cycles. Initial sprints focused on core functionalities such as user registration and login, dataset upload, and database integration. Later sprints included implementation of weather trend analysis, crop yield comparison, predictive insights, and report generation. The application was developed using Php in Visual Studio Code, while MySQL was used to manage datasets and analysis results. A modular architecture was followed to support scalability, maintainability, and future enhancements.

Testing Phase

Each component underwent thorough testing to ensure reliability, usability, and data accuracy. Testing types included:

- **Unit Testing** for individual modules such as authentication, dataset upload, chart generation, and report creation.
- **Integration Testing** to verify proper communication between the frontend (Php) and backend/database (MySQL).
- **User Interface Testing** across different screen sizes and web browsers.
- **User Acceptance Testing (UAT)** involving farmers, researchers, and administrators to validate real-world usability.

Deployment Phase

The system is planned to be deployed on a live production server using MySQL as the backend database and a cloud-based hosting environment. Before full public release, a staging version

will be deployed for a closed group of users to perform final beta testing. This allows real-world feedback from farmers, researchers, and administrators to refine system performance, usability, and analytical accuracy. After ensuring stability and resolving identified issues, the dashboard will be fully deployed for public use. Continuous monitoring tools and server analytics will support ongoing maintenance, updates, and performance tracking.

Advantages of Agile Methodology for This Project

- **Rapid Delivery and Feedback:** Sprint-based development enables quick implementation of features such as dataset upload, weather trend analysis, crop yield comparison, predictive insights, and report generation, allowing early user feedback.
- **Parallel Development:** Different modules (user dashboard, analytics engine, database integration, and administration panel) can be developed simultaneously, reducing overall development time.
- **Risk Reduction:** Continuous testing in each sprint helps identify and resolve issues early, ensuring system reliability, accuracy, and stability.
- **Collaboration:** Frequent feedback from farmers, researchers, and agricultural experts ensures the system meets real-world agricultural analysis and crop management needs.



Figure 2.1 Agile Activities

2.2 Tools & Technologies

The Weather Impact Analysis on Crops Dashboard leverages a modern web development stack to deliver a responsive, secure, and user-friendly platform for farmers, researchers, agricultural experts, and administrators. Built using Php and MySQL, the system integrates interactive user interfaces, backend services, and data visualization technologies to ensure smooth functionality, analytical accuracy, and scalability. The following outlines the tools and technologies used across key components of development.

Frontend Development

1. HTML5

- Provides the structural foundation of the web application.
- Organizes dashboard layouts, forms, tables, and content sections.

2. CSS3

- Used to style the application and create a visually appealing and responsive interface.
- Ensures compatibility across desktops, laptops, tablets, and mobile browsers.

3. JavaScript (ES6+)

- Adds interactivity to the dashboard.
- Supports dynamic features such as navigation, form validation, and chart updates.

4. Chart.js / Plotly

- Data visualization libraries used to generate interactive charts and graphs.
- Displays weather trends, crop yield comparisons, and predictive insights.

Backend Development

1. Python

- Core programming language used for data processing, analytics, and backend logic.

- Handles dataset cleaning, weather analysis, crop yield calculations, and report generation.

2. Php

- Lightweight PHP-based backend framework used to develop REST APIs and server-side services.
- Connects the frontend with the database and analysis modules.

3. MySQL

- Relational database management system used to store user accounts, weather records, crop datasets, analytical results, and reports.
- Ensures secure and efficient data storage.

4. Pandas

- Used for data cleaning, transformation, and statistical analysis.

5. NumPy

- Supports mathematical and numerical computations.

6. Scikit-learn

- Used for predictive modeling and crop yield forecasting.

2.3 Design

The design of the Weather Impact Analysis on Crops Dashboard outlines the system architecture, user workflows, and data management approach to ensure a scalable, secure, and user-friendly web platform. Developed using Php and MySQL, the design focuses on role-based access control, data visualization, and modular components to improve usability across different devices. The system supports three main roles: farmers, researchers/agricultural experts, and administrators. The following subsections include use case diagrams, user interaction scenarios, sequence diagrams, class diagrams, data flow diagrams, entity-relationship diagrams, database structure, and overall system architecture.

2.3.1 Use Case Diagrams

Actors:

- 1) **Farmer:** The primary user who uploads datasets, views weather trends, compares crop yields, accesses predictive insights, and downloads reports.
- 2) **Researcher / Agricultural Expert:** Analyzes weather and crop relationships, validates results, and generates recommendations.
- 3) **Administrator:** Manages user accounts, datasets, reports, system monitoring, and overall security.
- 4) **Guest:** A non-registered visitor who can view general project information but cannot access analytical features.

Main Use Cases (Based on Weather Impact Analysis on Crops Dashboard Features):

1. **Register / Login (Farmer / Researcher / Administrator):** Allows users to securely create accounts and access the system based on their roles.
2. **View and Update Profile (All Roles):** Enables users to view and edit personal and professional information.
3. **Upload Weather and Crop Datasets (Farmer / Researcher):** Allows users to import agricultural and meteorological data for analysis.
4. **Analyze Weather Trends (Farmer / Researcher):** Enables visualization of temperature, rainfall, humidity, and other environmental factors.
5. **Compare Crop Yields (Farmer / Researcher):** Allows users to evaluate crop performance under different weather conditions.
6. **Generate Predictive Insights (Researcher):** Provides forecasting and recommendations based on historical data.
7. **Download Reports (Farmer / Researcher):** Enables users to export analytical summaries and visual reports.
8. **Access Historical Analysis (Farmer / Researcher):** Lets users review previously generated results and datasets.

9. **Manage Users, Datasets, and Reports (Administrator):** Enables administrative control over system data and accounts.
10. **Monitor System Activity and Analytics (Administrator):** Allows administrators to track system usage and generate operational reports

Use Case Diagram:

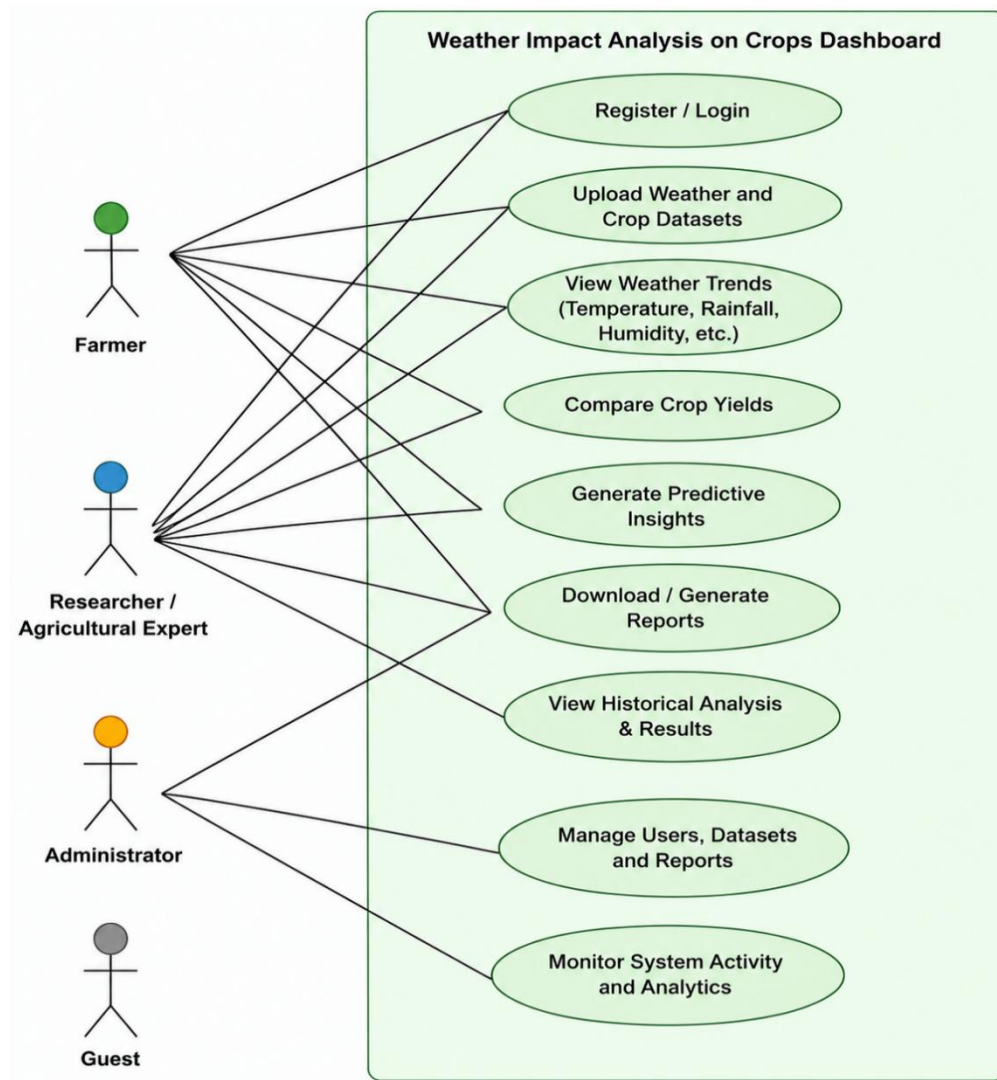


Figure 2.2 Use Case Diagram

2.3.2 Usage Scenario

Table 2. 1: Register User

Use Case Title	Register User
Use Case ID	1
Requirement ID	101
Description	- Allows a farmer, researcher, or administrator to create an account and access the Weather Impact Analysis on Crops Dashboard.
Pre-Conditions	- User has access to the web-based dashboard through a modern browser and a stable internet connection

Table 2. 2: Login

Use Case Title	Login (Farmer / Researcher / Administrator)
Use Case ID	2
Requirement ID	102
Description	- Allows a farmer, researcher, or administrator to securely log in to the Weather Impact Analysis on Crops Dashboard using a valid account.
Pre-Conditions	- User has a valid, activated account. System is operational.

Table 2. 3: Upload Dataset

Use Case Title	Upload Weather and Crop Dataset
Use Case ID	3
Requirement ID	103
Description	- Allows farmers and researchers to upload weather and crop datasets containing parameters such as temperature, rainfall, humidity, and crop yield.
Pre-Conditions	- System is online. User is logged in as Farmer or Researcher. Dataset file is available in supported format (CSV or Excel).

Table 2. 4: Generate Crop Recommendations

Use Case Title	Generate Crop Recommendations
Use Case ID	4
Requirement ID	104
Description	- Provides personalized crop recommendations and predictive insights based on uploaded weather and crop data.
Pre-Conditions	- User is logged in. Dataset has been uploaded successfully. Analysis has been completed. System is online.

2.3.3 Sequence Diagram:

The **Sequence Diagram** describes the step-by-step interaction between users and system components over time. It demonstrates how actions such as uploading datasets, performing analysis, and generating reports are executed through a sequence of messages.

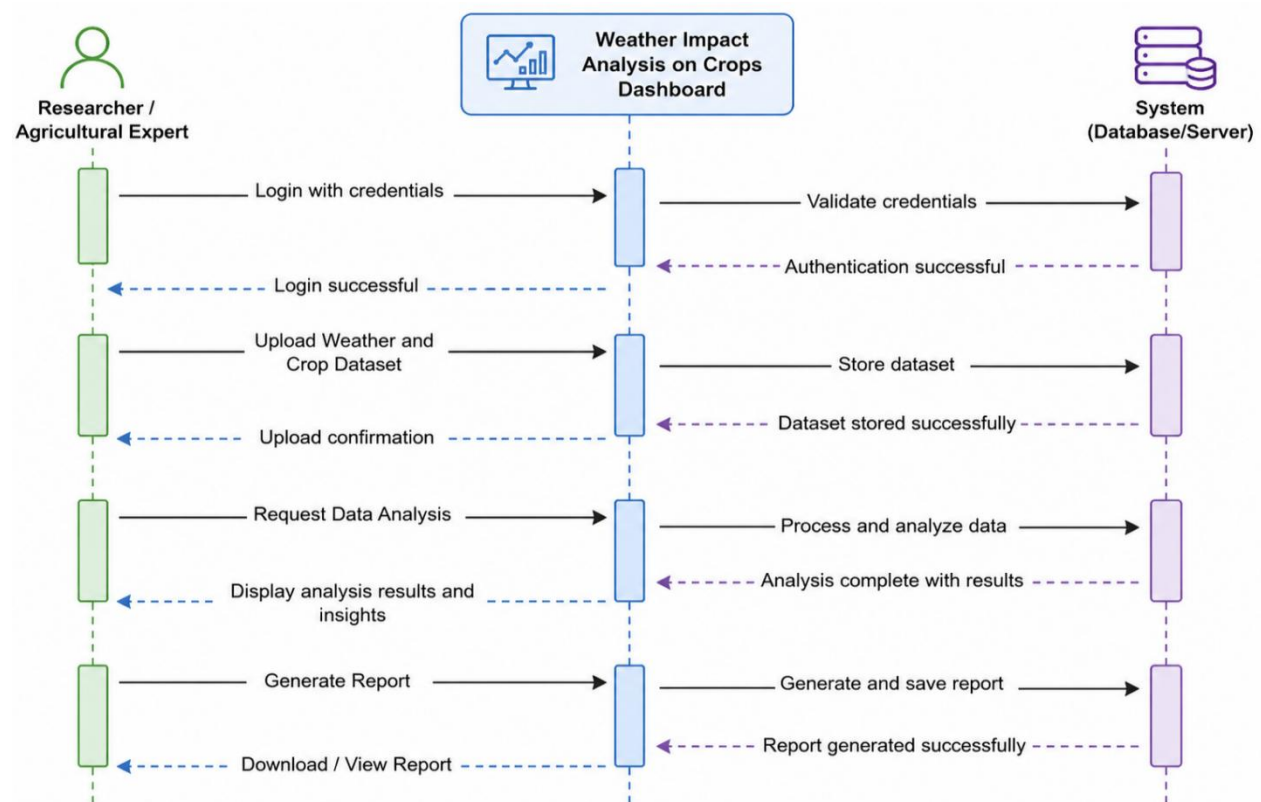


Figure 2.1 Sequence Diagram

2.3.4 Class Diagram:

The **Class Diagram** represents the static structure of the Weather Impact Analysis on Crops Dashboard by showing classes such as User, Farmer, Researcher, Weather Data, Crop Data, and Report, along with their attributes, methods, and relationships. It helps visualize how system components interact and manage agricultural information.

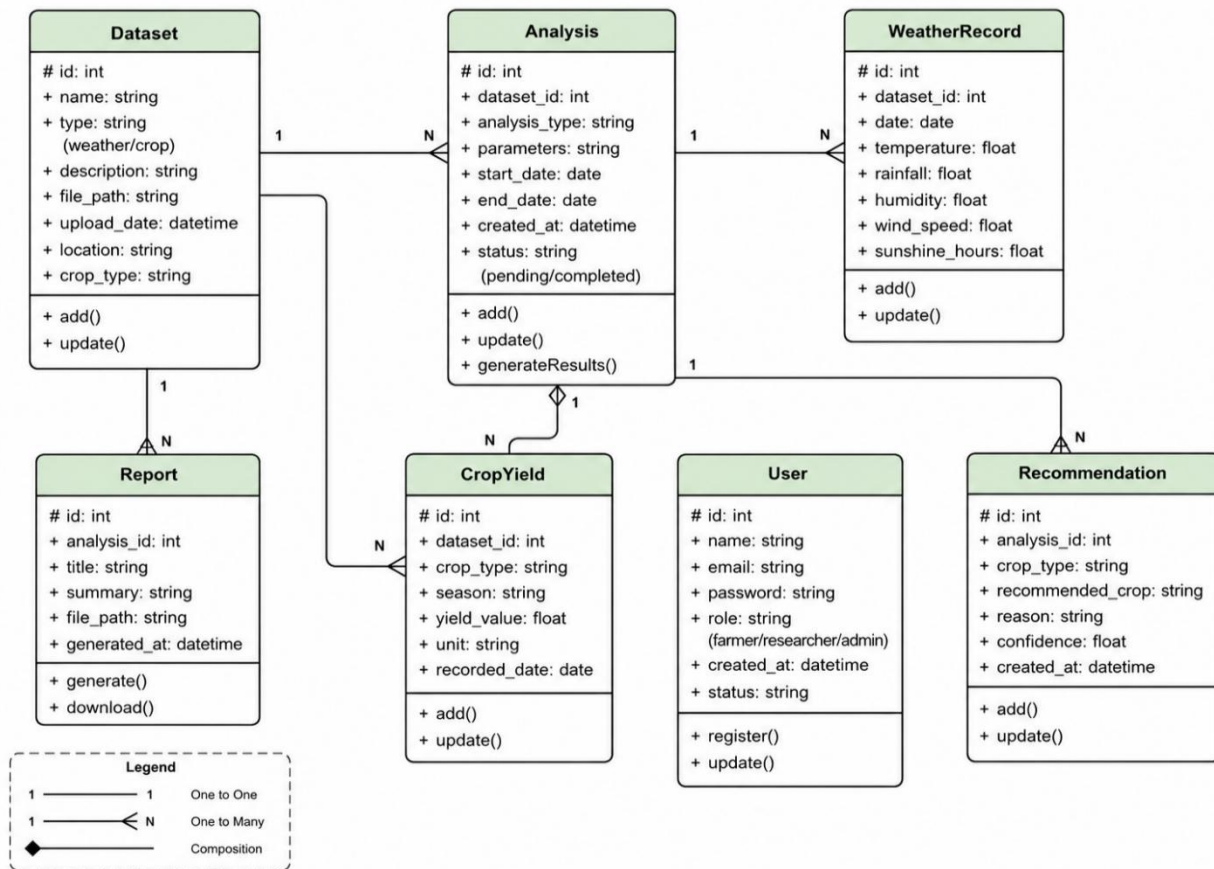


Figure 2.2 Class Diagram

2.3.4 Data Flow Diagram :

The **DFD** illustrates how data moves through the system, from dataset uploads by users to weather analysis, crop analysis, predictive insights, and report generation. It shows the interaction between users, processes, and databases within the dashboard.

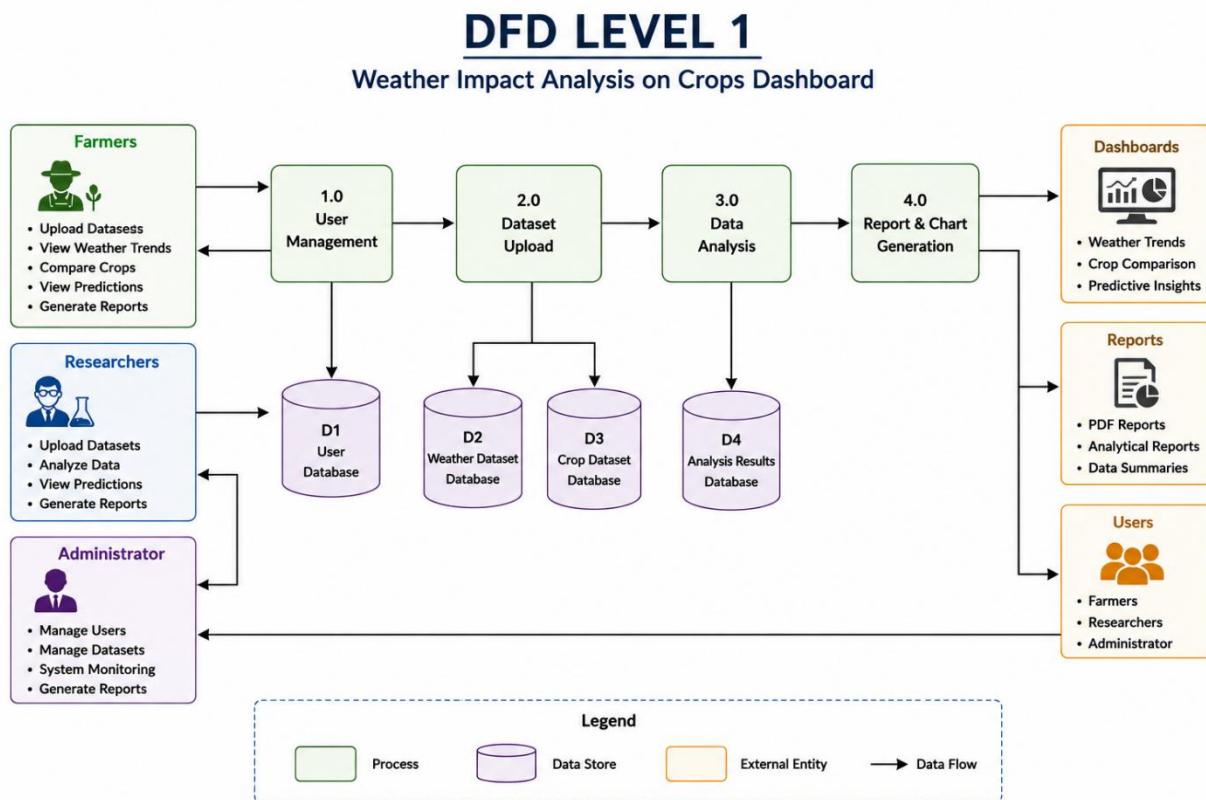
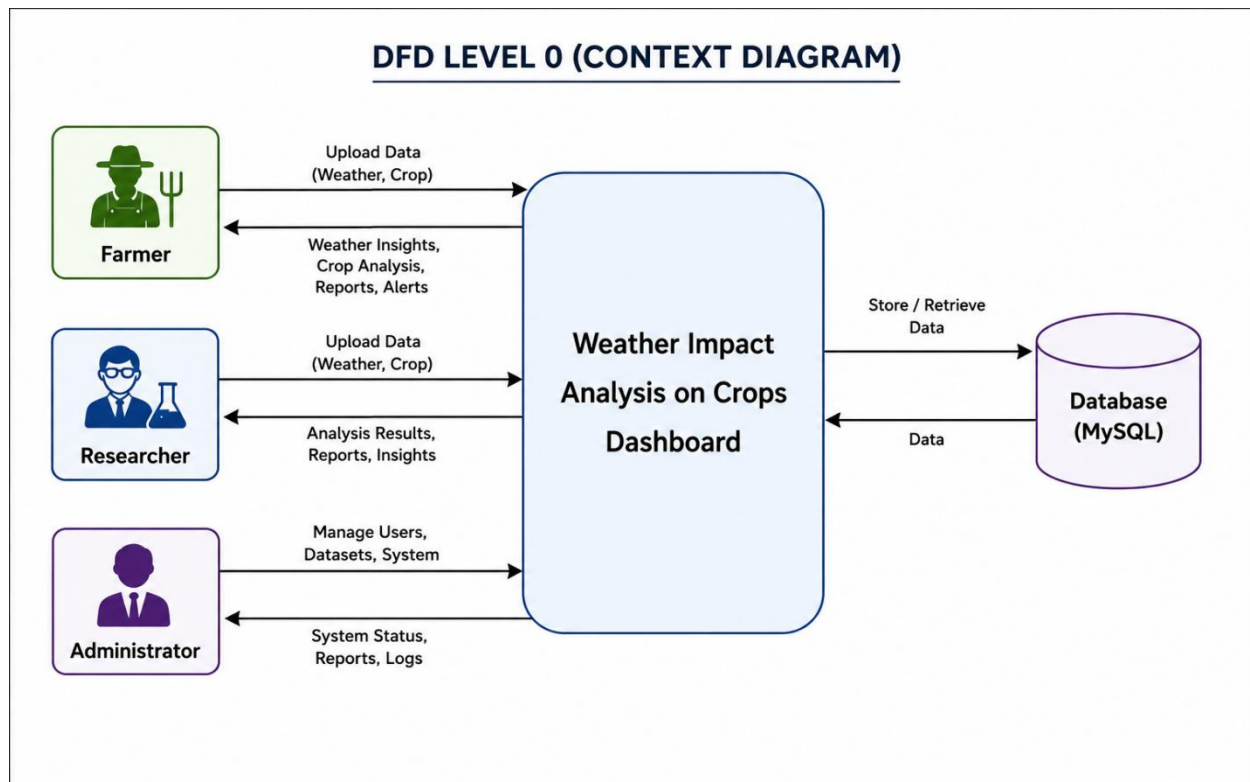


Figure 2.3 : DFD

2.3.5 ER Diagram:

The **ER Diagram** defines the database structure by showing entities such as Users, Weather Data, Crop Data, Analysis Results, and Reports, along with their relationships. It helps in organizing and managing data efficiently within the MySQL database.

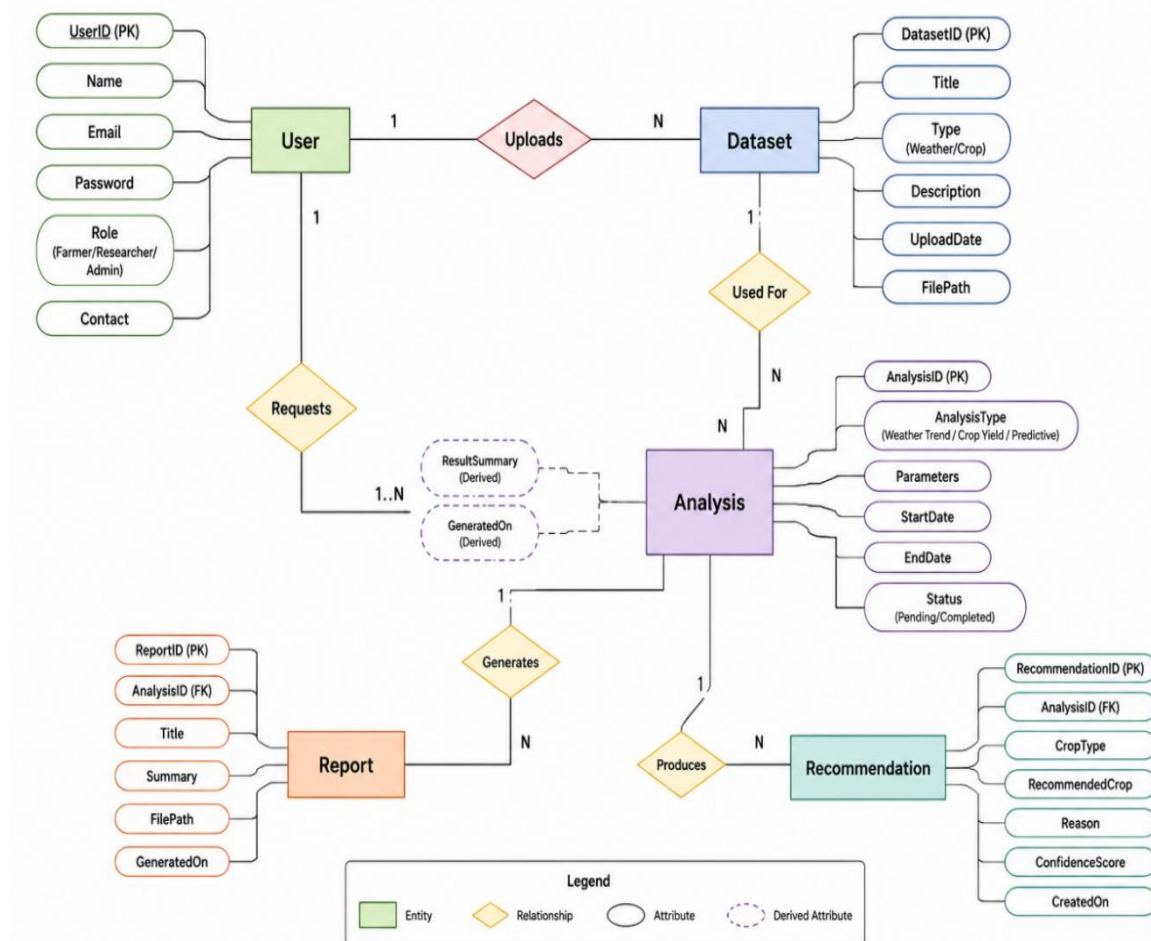


Figure 2.7 Entity Relationship Diagram

2.3.6 Database Model:

The **Database Model** provides the logical design of the system database, including tables, fields, primary keys, and foreign keys. It ensures secure storage, retrieval, and management of weather, crop, user, and analysis data required by the dashboard.

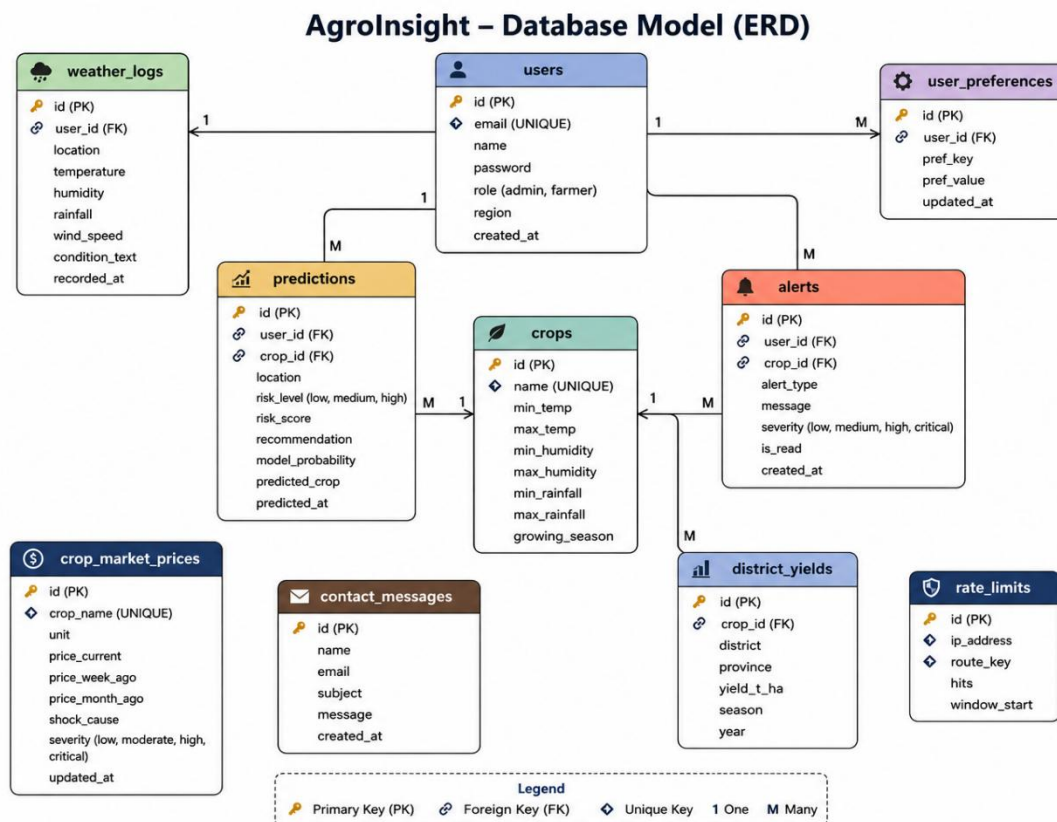


Figure 2.8 Database Model 1

2.3.7 Architecture:

The Weather Impact Analysis on Crops Dashboard adopts a 3-Tier Architecture, a modular and scalable design that separates responsibilities to ensure maintainability, flexibility, and performance. This architecture is highly suitable for web-based agricultural analytics systems developed using Python and MySQL, providing role-specific functionality for farmers, researchers, and administrators. The architecture consists of the Presentation Tier, Application Tier, and Data Tier, where each layer manages a specific part of the system workflow and communicates through secure and well-defined interfaces.

Presentation Tier

This tier represents the user interface of the web application, developed using HTML, CSS, JavaScript, and responsive dashboard components. It provides a user-friendly and visually appealing experience for all users. Farmers interact with screens to register, log in, upload

weather and crop datasets, visualize trends, compare crop yields, and download reports. Researchers use the dashboard to perform advanced analysis, generate predictive insights, and create recommendations. Administrators access a dedicated control panel to monitor users, datasets, and system performance. The Presentation Tier communicates user requests to the Application Tier through secure APIs and displays analytical results efficiently, ensuring accessibility and smooth performance across desktops, laptops, tablets, and mobile devices.

Application Tier

The core logic and functionality are handled in this middle layer, developed using Python with frameworks such as Flask or Django. This tier processes user input, manages authentication sessions, performs role-specific operations, and controls the overall workflow of the dashboard. Farmers can upload datasets and view analysis results, while researchers can generate forecasts and recommendations. Administrators can manage users and system configurations. Business logic for data preprocessing, weather trend analysis, crop yield comparison, predictive modeling, report generation, and notifications is centralized in this layer to improve efficiency and reduce errors. Integration with analytical libraries such as Pandas, NumPy, and Scikit-learn is also managed within this tier.

Data Tier

The backend data storage is managed through MySQL, which stores structured information related to users, uploaded datasets, weather records, crop details, analysis results, recommendations, and generated reports. The database supports efficient relationships between weather parameters and crop performance, enabling quick retrieval of historical trends and predictive insights. The system ensures secure data handling through authenticated access, validation rules, and role-based permissions. Regular backups, controlled database privileges, and secure access mechanisms are implemented to maintain data availability, integrity, and confidentiality.

The 3-Tier Architecture ensures modular development, easy maintenance, and better scalability of the Weather Impact Analysis on Crops Dashboard. The separation of interface, logic, and data improves system performance, security, and reliability. It also supports future enhancements

such as real-time weather API integration, satellite data analysis, AI-based crop recommendations, and advanced agricultural analytics.

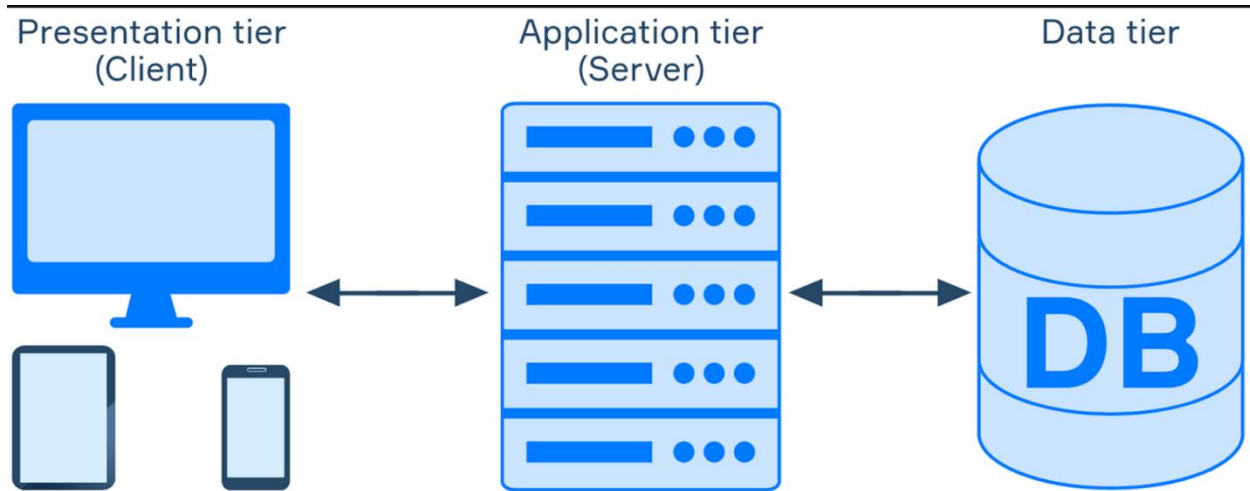


Figure 2.8 Application Architecture

Chapter 3 - RESULTS & DISCUSSION

3.1 Testing:

Testing for the Weather Impact Analysis on Crops Dashboard was conducted at multiple levels to ensure a reliable, secure, and user-friendly experience for farmers, researchers, and administrators. The testing process focused on validating core features such as user registration, login authentication, dataset upload, weather trend analysis, crop yield comparison, predictive modeling, recommendation generation, and report creation. The application was developed using Python in Visual Studio Code and uses MySQL for data storage. The following section outlines the testing approaches, tools, and results achieved during system evaluation.

1. Unit Testing

- **Purpose:** Validate individual components such as user registration, login authentication, dataset upload, weather analysis, and report generation.
- **Procedure:** Developers performed unit testing on Python functions and frontend components to verify application logic such as authentication, file validation, data preprocessing, chart generation, and role-based access for farmers, researchers, and administrators. Edge cases such as invalid login credentials, unsupported file formats, empty input fields, and missing dataset columns were also tested.
- **Tools:** Python, Flask/Django, Pandas, Visual Studio Code.
- **Example Test Case:** Verify that the dataset upload function rejects files missing required columns such as temperature, rainfall, humidity, or crop yield and displays an appropriate validation message.
- **Outcome:** Achieved stable component functionality with most input validation and processing issues identified and resolved during early development stages.

2. Integration Testing

- **Purpose:** Ensure smooth interaction between application modules such as the web frontend, Python backend services, MySQL database, and analytical components.

- **Procedure:** Integration testing was performed to verify workflows such as user registration, login authentication, dataset upload, weather analysis, crop comparison, recommendation generation, and report creation. Data synchronization between frontend interfaces and backend processing modules was also validated.
- **Tools:** Visual Studio Code, browser developer tools, MySQL Workbench.
- **Example Test Case:** Confirm that uploading a dataset successfully stores file metadata in MySQL, triggers the analysis process, and displays generated charts and insights on the dashboard.
- **Outcome:** Minor issues related to data mapping and validation were identified and resolved, improving communication between frontend and backend components.

3. System Testing

- **Purpose:** Test the complete Weather Impact Analysis on Crops Dashboard to ensure all functionalities work correctly as an integrated system.
- **Procedure:** Scenarios such as user registration, dataset upload, weather trend analysis, crop yield comparison, predictive modeling, recommendation generation, and report downloading were tested in a controlled environment.
- **Tools:** Visual Studio Code, modern web browsers, MySQL Workbench.
- **Example Test Case:** A farmer logs into the system, uploads a crop dataset, performs analysis, receives recommendations, and downloads a summary report.
- **Outcome:** All major use case flows operated successfully. Minor UI and validation issues were identified and corrected to improve overall usability and system stability.

4. Performance Testing

- **Purpose:** Validate the dashboard's performance under multiple user requests and large dataset operations.
- **Procedure:** Simulated multiple users uploading datasets and executing analyses simultaneously to test response times, database operations, and memory usage.
- **Tools:** Browser developer tools, MySQL monitoring tools, Python profiling utilities.

- **Example Test Case:** Confirm that weather charts and crop analysis results load within a few seconds when multiple users access the system concurrently.
- **Outcome:** The application maintained stable performance with acceptable response times. Database indexing and optimized processing improved loading speed and responsiveness.

5. Usability Testing

- **Purpose:** Evaluate how user-friendly and accessible the dashboard is for farmers and researchers.
- **Procedure:** Participants tested key features such as registration, login, dataset upload, analysis, recommendation viewing, and report downloading. Feedback regarding navigation, interface design, and usability was collected after testing sessions.
- **Tools:** Google Forms, Figma, Visual Studio Code.
- **Example Test Case:** A farmer uploads a dataset, views weather impact charts, and downloads recommendations without external assistance.
- **Outcome:** Users found the dashboard easy to navigate and understand. Minor improvements were made to chart labels, button placement, and validation messages to enhance the user experience.

Testing Coverage

Testing covered the following functionalities:

- Farmer, researcher, and administrator registration/login
- Weather and crop dataset upload
- Weather trend analysis and visualization
- Crop yield comparison
- Predictive modeling and recommendation generation
- Report creation and downloading

Best Practices for Test Cases

- Used descriptive test case IDs for better traceability

- Aligned test cases with system functional requirements
- Included both valid and invalid input scenarios
- Maintained modular test cases for reusability
- Used realistic agricultural datasets in staging for accurate feedback

By adhering to these practices, the Weather Impact Analysis on Crops Dashboard development process ensured that each critical module remained reliable, secure, and responsive to user requirements while also supporting future scalability and advanced analytical enhancements.

3.2 Test Cases:

Test Case 1 – User Registration

Test ID	TC-01
Feature	User Registration
Description	Validate new user registration with valid and invalid data
Priority	High
Estimated Time	1 Hour
Test Data	Name, Email, Password, Role
Input	Blank fields / Valid data
Expected Output	Show error or create account
Actual Output	Error displayed / Account created
Status	Pass

Test Case 2 – User Login

Test ID	TC-02
Feature	User Login
Description	Validate that registered users can log into the system using valid credentials and are restricted with invalid credentials
Priority	High
Estimated Time	1 Hour
Test Data	User Email, Password
Input	Valid / Invalid login credentials
Expected Output	Successful login and dashboard access OR error message for invalid login.
Actual Output	User redirected to dashboard / Error message displayed
Status	Pass

Test Case 3 – Dataset Upload

Test ID	TC-03
Feature	Dataset Upload

Description	Ensure users can upload weather and crop datasets successfully
Priority	High
Estimated Time	1 Hour
Test Data	CSV/Excel file containing weather and crop data
Input	Select valid or invalid dataset file
Expected Output	Dataset uploaded successfully or validation error displayed
Actual Output	Upload completed / Error displayed
Status	Pass

Test Case 4 – Weather Trend Analysis

Test ID	Weather Trend Analysis
Feature	Validate that the system generates charts for temperature, rainfall, humidity, and weather parameters
Description	High
Priority	1 Hour
Estimated Time	Uploaded weather dataset
Test Data	Select analysis parameters and run analysis
Input	Interactive weather charts displayed
Expected Output	Charts generated successfully
Actual Output	Weather Trend Analysis
Status	Validate that the system generates charts for temperature, rainfall, humidity, and weather parameters

Test Case 5 – Crop Yield Comparison

Test ID	TC-05
Feature	Crop Yield Comparison
Description	Validate that the system compares crop yields under different weather conditions
Priority	High
Estimated Time	1 Hour
Test Data	Crop names, yield records, weather data
Input	Select crops and comparison criteria
Expected Output	Comparative charts and summaries displayed
Actual Output	Comparison generated successfully
Status	Pass

Test Case 6 – Recommendation Generation

Test ID	TC-06
Feature	Recommendation Generation
Description	Validate that the system generates crop recommendations based on weather and

	analysis
Priority	High
Estimated Time	1 Hour
Test Data	Analysis results and predictive model output
Input	Run recommendation module
Expected Output	Personalized recommendations displayed
Actual Output	Recommendations generated successfully
Status	Pass

Test Case 7 – Report Generation

Test ID	TC-07
Feature	Report Generation
Description	Validate that users can generate and download analytical reports
Priority	High
Estimated Time	1 hour
Test Data	Weather analysis and recommendation results
Input	Click "Generate Report"
Expected Output	Report created and available for download
Actual Output	Report generated successfully
Status	Pass

Test Case 8 – Notification System

Test ID	TC-8
Feature	Notification System
Description	Validate that users receive notifications for completed analyses, uploaded datasets, and generated reports
Priority	High
Estimated Time	1 Hour
Test Data	Dataset upload and report generation events
Input	Trigger system events
Expected Output	Notification displayed to the user
Actual Output	Notification displayed successfully
Status	Pass

Test Case 9 – User Profile Management

Test ID	TC-9
Feature	User Profile Management
Description	Validate that User can view and update their profile information
Priority	Medium

Estimated Time	1 Hour
Test Data	Name, Email, Contact Number, Profile Picture
Input	Edit and save profile details
Expected Output	Profile updated successfully and changes reflected in account
Actual Output	Profile updated / Error displayed for invalid data
Status	Pass

Test Case 10 – Historical Analysis Access

Test ID	TC-10
Feature	Historical Analysis Access
Description	Validate that users can access and review previous analyses and reports
Priority	High
Estimated Time	1 Hour
Test Data	Stored analysis results
Input	Select previous analysis
Expected Output	Historical results displayed correctly
Actual Output	Analysis shown successfully
Status	Pass

Test Case 11 – Admin Dashboard Management

Test ID	TC-11
Feature	Admin Dashboard Management
Description	Validate that admin can view and manage overall system data including users, doctors, appointments, and system activities
Priority	high
Estimated Time	1 Hour
Test Data	User records, dataset records, report data
Input	Login as admin and access dashboard
Expected Output	Dashboard loads with complete system overview and management options
Actual Output	Dashboard displayed successfully with all data
Status	Pass

3.3 Conclusion:

The Weather Impact Analysis on Crops Dashboard was developed to streamline the analysis of weather conditions and their effects on crop productivity through a digital, efficient, and user-friendly system. The project focused on key features such as dataset upload, weather trend visualization, crop yield comparison, predictive modeling, recommendation generation, report creation, and role-based dashboards.

Throughout the development process, a modular and scalable architecture was implemented to ensure easy maintenance and future enhancements, such as real-time weather API integration, satellite data analysis, AI-based crop recommendations, and advanced agricultural analytics. The application supports multiple roles (farmers, researchers, and administrators) with dedicated dashboards to ensure smooth interaction and effective workflow management.

The system was designed to be responsive, secure, and user-friendly. Key challenges included processing large datasets, maintaining data consistency, generating accurate predictive insights, and ensuring smooth role-based access control. These challenges were resolved through iterative development, comprehensive testing, and continuous improvements.

In the future, the Weather Impact Analysis on Crops Dashboard can be extended into a complete smart agriculture platform with features such as real-time weather alerts, automated crop advisory notifications, integration with IoT sensors, and AI-driven decision support, making agricultural planning more accessible, accurate, and efficient, especially for farmers in remote and climate-sensitive regions.

Chapter 4 - USER MANUAL

Login Page :

The Login Page allows registered users to securely access the system by entering their username/email and password. It authenticates user credentials and provides access to features based on their assigned role and permissions.

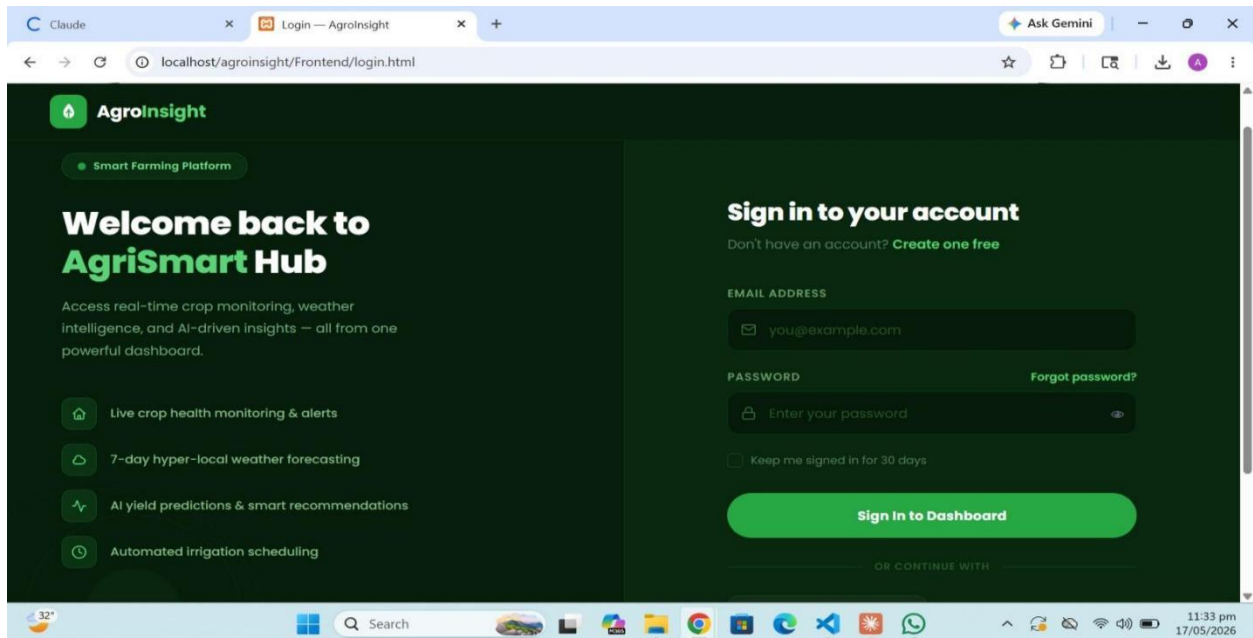


Figure 4.1 Login page

Home Page :

Serves as the landing page of the system, providing an overview of AgroInsight, its features, and benefits for farmers.

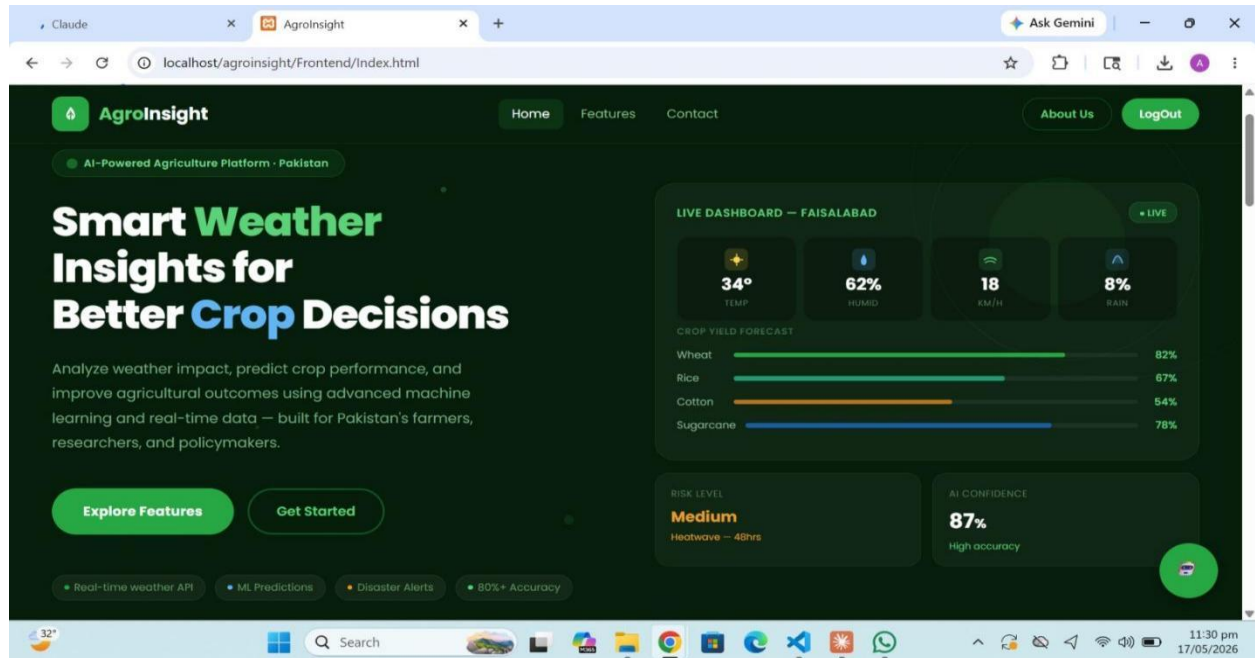


Figure 4.2 Admin logged in

Admin Dashboard :

Displays key information such as weather conditions, crop recommendations, risk analysis, and system statistics in a visual format.

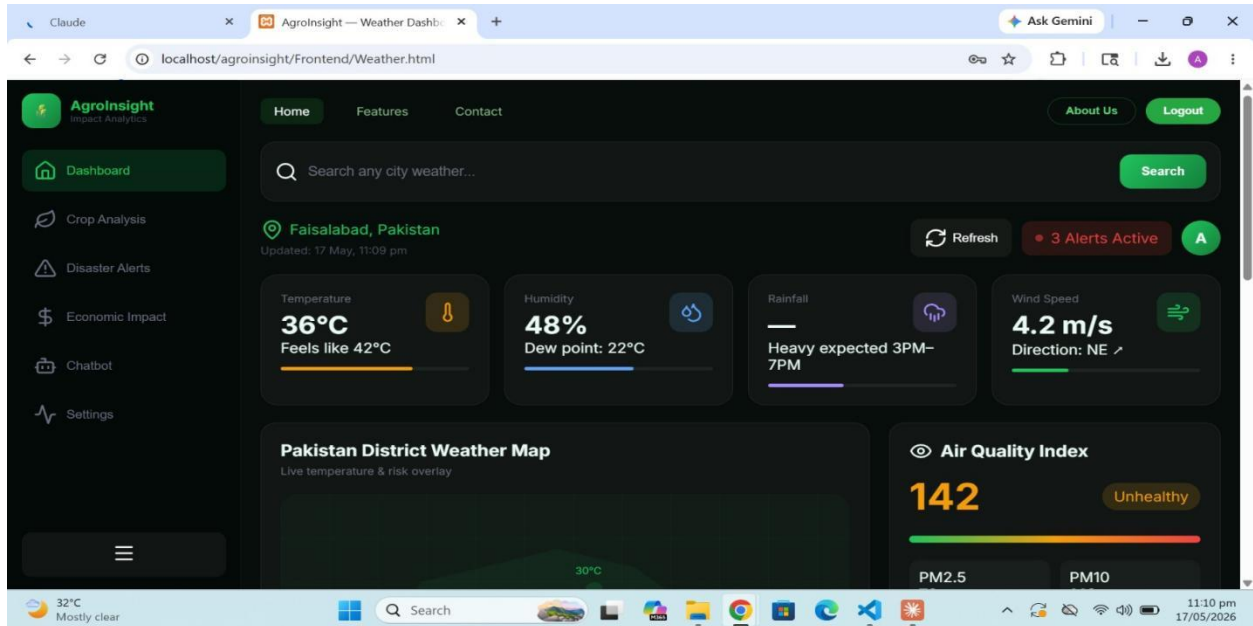


Figure 4.3 Dashboard of Admin

Crop-Wise Impact Analysis :

Allows users to enter soil and weather parameters and receive recommendations for the most suitable crops.

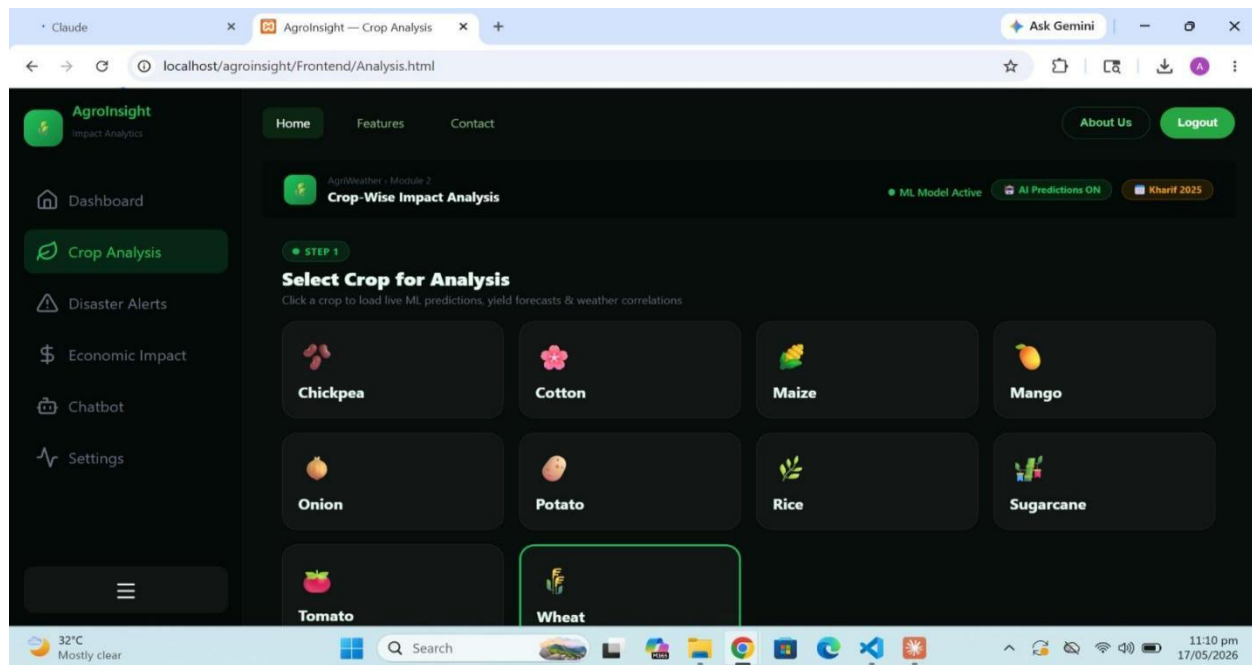


Figure 4.3 Crop-wise impact analysis

Disaster Alert :

Displays important alerts, warnings, and recommendations related to weather changes and crop management.

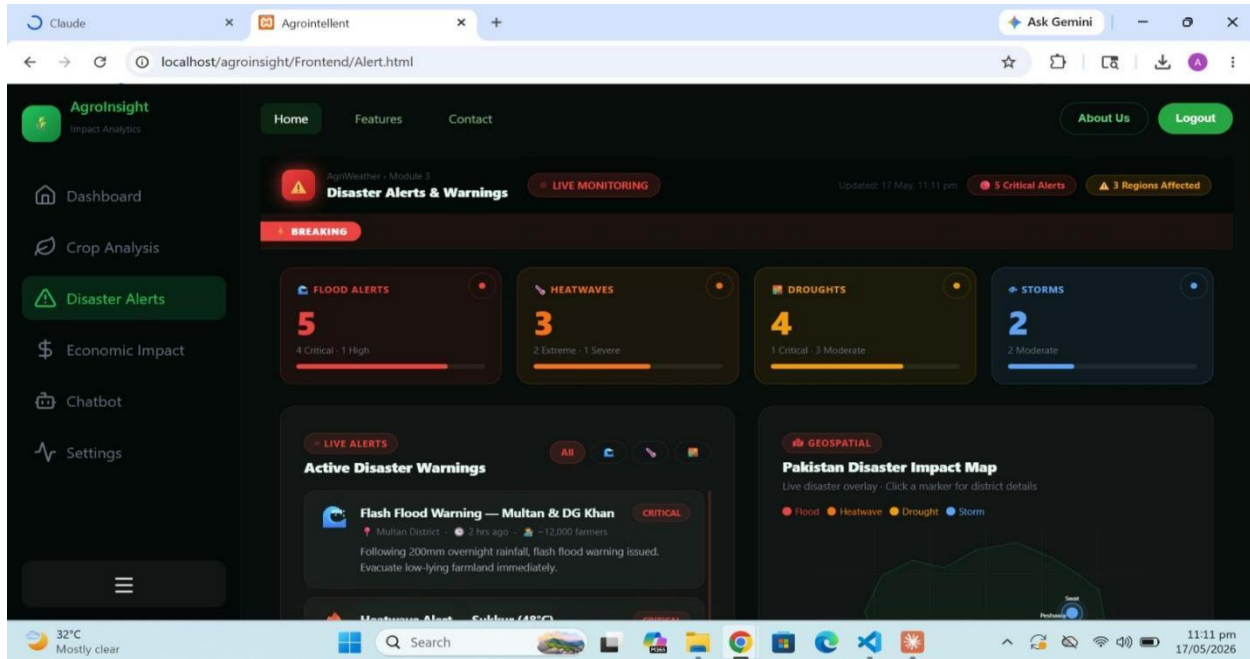
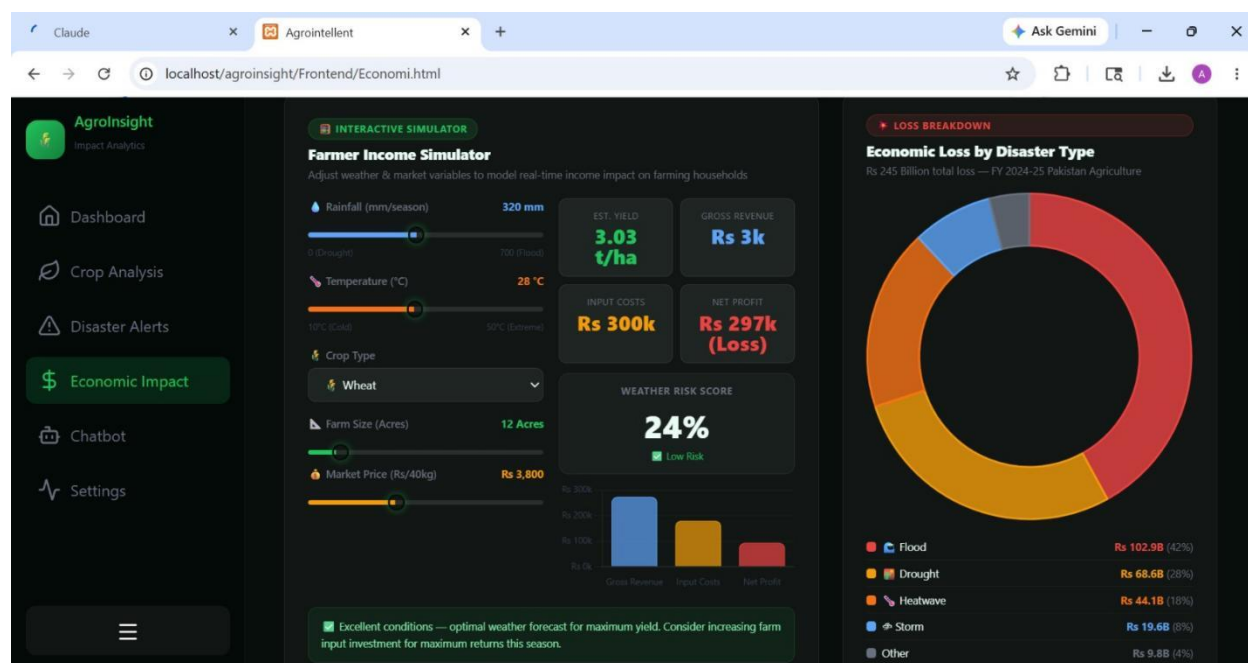


Figure 4.3 Disaster alert

Economic Analysis :

The economic analysis evaluates the financial feasibility and benefits of the AgroInsight system. The project is designed to provide farmers with accurate crop recommendations and weather-based insights, helping reduce losses and improve agricultural productivity



Future Worth

The Weather Impact Analysis on Crops Dashboard demonstrates strong future potential in improving digital agriculture through efficient weather-crop analysis and centralized agricultural data management. By simplifying dataset upload, weather trend visualization, crop yield comparison, predictive modeling, recommendation generation, report creation, and notifications, the system enhances agricultural decision-making and operational efficiency.

The current application supports secure role-based access for farmers, researchers, and administrators while maintaining reliable performance and structured data management. Future enhancements may include AI-based crop recommendations, real-time weather alerts, satellite image integration, automated disease risk prediction, and smart crop planning.

The scalable architecture developed using Python, Visual Studio Code, and MySQL provides a strong foundation for integrating advanced agricultural technologies and cloud-based services.

Additional features such as multilingual support, IoT sensor integration, drone data analysis, and advanced agricultural analytics can further improve the system's effectiveness.

Through continuous development, user feedback, and modern agricultural technologies, the Weather Impact Analysis on Crops Dashboard has the potential to evolve into a complete smart agriculture solution that improves crop productivity, reduces climate-related risks, and enhances farming efficiency in the future.

References

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- 2) [Software Engineering by Ian Sommerville](#)
- 3) [Python Official Documentation](#)
- 4) [Flask Documentation](#)
- 5) [MySQL Documentation](#)
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- 7) [Design Patterns: Elements of Reusable Object-Oriented Software](#)